IT24101011 – Jayaweera A.J.D
Assigned Model: K-Nearest Neighbours (KNN)

1. Individual Role and Contributions

- My main responsibility in the GrainWise project was developing and evaluating the K-Nearest Neighbours (KNN) classification model to distinguish Cammeo and Osmancik rice varieties using their physical measurements.
- I worked on the KNN.ipynb notebook, implementing model training, hyperparameter tuning, repeated cross-validation, seed stability testing, and sensitivity analysis. I also investigated the effect of feature scaling on KNN performance and saved the evaluation results for comparison with the models developed by my teammates.

2. Technical Learning and Key Decisions

- One of my most important learning experiences was understanding how KNN classifies new observations based on the distances to neighbouring training samples. I learned that feature scaling is especially important for distance-based algorithms because measurements with larger numerical values can dominate distance calculations.
- I used GridSearchCV to evaluate 20 hyperparameter combinations, including neighbour counts, distance weighting, and Manhattan and Euclidean distance metrics. The selected configuration used 25 neighbours, Euclidean distance ($p = 2$), and distance-based weighting.
- Using repeated stratified cross-validation, my model achieved 92.80% accuracy, 91.59% F1-score, and 97.59% ROC-AUC. These results helped me understand how appropriate hyperparameter selection and validation contribute to reliable classification performance.

3. Challenges and Problem-Solving

- A major challenge was understanding how differences in feature scales affected the KNN model. To investigate this, I evaluated the same model with and without standardisation. The accuracy improved from 88.87% without scaling to 92.80% with scaling, demonstrating a 3.94 percentage-point improvement.
- Another challenge was selecting an appropriate number of neighbours. I performed a sensitivity analysis using different values of k from 1 to 31 to examine how neighbourhood size influenced classification accuracy. This helped me understand the relationship between model flexibility, sensitivity to individual observations, and generalisation.

4. Critical Evaluation and Reflection

- Comparing KNN with Logistic Regression, SVM, and Random Forest helped me understand that different algorithms can achieve similar predictive performance despite using different learning approaches.
- I also evaluated KNN using different random seeds (42, 7, and 2024), obtaining consistent accuracy results. This strengthened my understanding of model stability and the importance of reliable evaluation.
- Although KNN performed competitively, I recognised its limitations, particularly its dependence on feature scaling, prediction-time distance calculations, and limited interpretability compared with models such as Logistic Regression.

5. Personal Growth and Future Improvements

- This project improved my practical understanding of distance-based learning, preprocessing, hyperparameter optimisation, cross-validation, and model comparison. Most importantly, I learned that appropriate preprocessing decisions can significantly influence performance, sometimes more than changing the algorithm itself.
- In future projects, I would explore feature selection, alternative distance metrics, and strategies for improving KNN efficiency when working with larger datasets.